BUSINESS ANALYTICS • WEEK 8

Macro-Economic Causal Modeling with VAR Analytics

Business Analytics – Week 8 Task
GDP • Inflation • Interest Rates
Python-based time-series analysis

45-SLIDE PRESENTATION

Introduction

- Macroeconomic indicators influence business activity and market conditions.
- GDP represents economic output, inflation measures price changes, and interest rates represent borrowing costs.
- VAR analysis studies how these indicators move together over time.

Project Overview

- Collect multi-decade macroeconomic time-series data.
- Prepare and transform the indicators.
- Test stationarity using statistical methods.
- Select the best lag length using Akaike Information Criterion (AIC).
- Fit a Vector Autoregression (VAR) model and study impulse responses.

Problem Statement

- Economic indicators are connected and may influence one another.
- A change in interest rates may affect inflation and GDP growth.
- The project uses VAR analytics to study these dynamic relationships.
- The results can support corporate planning and market-risk discussions.

Objectives

- Analyze long-term GDP, inflation and interest-rate data.
- Check whether the selected series are stationary.
- Choose an appropriate lag configuration using AIC.
- Estimate a multivariate VAR model.
- Use impulse-response analysis and forecasts to interpret trends.

Selected Economic Indicators

- Real GDP – measures inflation-adjusted economic output.
- Inflation – measures the rate at which prices increase.
- Federal Funds Rate – represents a key short-term policy interest rate.
- These indicators are used as an example dataset for VAR analysis.

Why Use Multi-Decade Data?

- Long time periods include different business-cycle conditions.
- The data may contain expansions, slowdowns and policy changes.
- More observations help analyze relationships across time.
- The selected period should be consistent and documented.

What is VAR?

- VAR means Vector Autoregression.
- It models multiple time-series variables together.
- Each variable is explained using past values of all variables in the system.
- VAR is useful when variables may influence one another over time.

VAR Model Structure

- A VAR model contains equations for every variable.
- Each equation uses lagged values of GDP growth, inflation and interest rates.
- The same lag length is generally used across equations.
- The model captures dynamic relationships, not automatic proof of causation.

Study Workflow

- 1. Download and inspect data.
- 2. Clean and align the dates.
- 3. Transform variables and test stationarity.
- 4. Select lag length using AIC.
- 5. Fit VAR, check diagnostics, analyze IRFs and forecast.

Data Sources

- Use Federal Reserve Economic Data (FRED).
- Real GDP series: GDPC1 (quarterly).
- Consumer Price Index: CPIAUCSL (monthly).
- Federal Funds Rate: FEDFUNDS (monthly).
- Record the source, frequency, units and download date.

Study Period and Frequency

- Example study period: 1985 Q1 to 2025 Q4.
- Use quarterly frequency for the VAR model.
- GDP is already quarterly.
- Convert monthly CPI and interest-rate data to quarterly averages.
- Use the same quarter index for all variables.

Tools and Libraries

- Python – programming language.
- pandas – data handling and date alignment.
- numpy – numerical calculations.
- matplotlib – charts.
- statsmodels – ADF test, VAR estimation and impulse responses.

Install Required Libraries

```
pip install pandas numpy matplotlib statsmodels pandas-datareader

Run this command in the terminal or notebook.
The libraries support data collection, testing, modeling and charts.
```

Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from pandas_datareader import data as pdr
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.api import VAR
```

Download GDP Data

```
start = '1985-01-01'
end = '2025-12-31'
gdp = pdr.DataReader('GDPC1', 'fred', start, end)
gdp = gdp.resample('QS').mean()
GDP is real gross domestic product, measured quarterly.
Check the downloaded date range before analysis.
```

Download CPI and Interest Rates

```
cpi = pdr.DataReader('CPIAUCSL', 'fred', start, end)
rate = pdr.DataReader('FEDFUNDS', 'fred', start, end)
cpi_q = cpi.resample('QS').mean()
rate_q = rate.resample('QS').mean()
Monthly series are converted to quarterly averages.
```

Combine and Clean Data

```
data = pd.concat([gdp, cpi_q, rate_q], axis=1)
data.columns = ['GDP', 'CPI', 'Interest_Rate']
data = data.dropna()
data = data.sort_index()
print(data.head())
print(data.isnull().sum())
```

Create Analysis Variables

```
data['GDP_Growth'] = 100 * np.log(data['GDP']).diff()
data['Inflation'] = 100 * np.log(data['CPI']).diff()
data['Interest_Rate'] = data['Interest_Rate']
var_data = data[['GDP_Growth', 'Inflation', 'Interest_Rate']].dropna()
Log differences express GDP growth and inflation approximately in percent.
```

Understand Stationarity

- A stationary series has a stable statistical pattern over time.
- Its mean and variance do not change systematically.
- Its autocovariance depends on the lag, not the date.
- Many VAR methods assume stationary input series.

Why Stationarity Matters

- Non-stationary series can produce misleading relationships.
- Trends may create high correlations even without a meaningful connection.
- Transformations such as differencing can help achieve stationarity.
- Test each model variable before fitting VAR.

Augmented Dickey-Fuller Test

- The ADF test checks for a unit root.
- Null hypothesis (H0): the series has a unit root.
- Alternative hypothesis (H1): the series is stationary.
- A p-value below 0.05 is commonly treated as evidence against H0.
- Interpret the result together with the time-series plot.

ADF Test: Python Function

```
def adf_test(series, name):
    result = adfuller(series.dropna(), autolag='AIC')
    print(name, 'ADF statistic:', result[0])
    print(name, 'p-value:', result[1])
    print('Critical values:', result[4])
    return result[1]
```

Run Stationarity Tests

```
for col in var_data.columns:
    adf_test(var_data[col], col)

Run the test for GDP growth, inflation and interest rate.
Record each ADF statistic, p-value and critical values.
```

Interpreting ADF Results

- If p-value < 0.05, reject the unit-root null hypothesis.
- If p-value ≥ 0.05, the test does not provide enough evidence of stationarity.
- Do not claim a series is stationary only because its chart looks stable.
- Use a suitable transformation and test again when needed.

Differencing Non-Stationary Series

- If a variable is non-stationary, consider first differencing.
- Example: change in interest rate = current rate minus previous rate.
- Apply transformations only when justified by tests and economic meaning.
- Retest the transformed series before VAR estimation.

Differencing: Python Example

```
var_stationary = var_data.copy()
var_stationary['Interest_Rate'] = var_stationary['Interest_Rate'].diff()
var_stationary = var_stationary.dropna()
adf_test(var_stationary['Interest_Rate'], 'Differenced Rate')

Use this only if the original interest-rate series fails the stationarity check.
```

Visualize the Series

- Plot GDP growth, inflation and interest rates over time.
- Look for trends, sudden changes and unusual observations.
- Compare the plot with the ADF test results.
- Visual inspection supports, but does not replace, statistical testing.

Plot the Variables

```
var_stationary.plot(subplots=True, figsize=(10, 7))
plt.tight_layout()
plt.show()

The plots help compare the movement of all model variables.
```

Choose Lag Length

- A lag is a previous time period used by the model.
- For quarterly data, lag 1 represents the previous quarter.
- Too few lags may miss relationships.
- Too many lags may use too many parameters.
- AIC helps compare candidate lag lengths.

Akaike Information Criterion

- AIC balances model fit and model complexity.
- A lower AIC indicates a preferred model among the candidates tested.
- AIC does not guarantee that a model is correct.
- Check residual diagnostics and economic reasonableness too.

Select Lag Using AIC

```
model = VAR(var_stationary)
lag_selection = model.select_order(maxlags=8)
print(lag_selection.summary())
best_lag = lag_selection.selected_orders['aic']
print('Selected lag by AIC:', best_lag)
```

Fit the VAR Model

```
var_result = model.fit(best_lag)
print(var_result.summary())

The fitted model estimates an equation for each variable.
Review coefficients, standard errors and model fit.
```

Model Diagnostics

- Check whether residuals behave like white noise.
- Use serial-correlation tests such as the VAR Portmanteau test.
- Inspect residual plots and stability conditions.
- A model with serious diagnostic problems should be reconsidered.

Stability Check

- VAR stability requires the roots of the companion matrix to lie inside the unit circle.
- A stable VAR supports standard impulse-response interpretation.
- Use the model's stability test before interpreting long-run responses.
- If unstable, revisit transformations and lag choice.

Impulse-Response Analysis

- An impulse response traces how variables react over future periods after a shock.
- Example: examine how GDP growth responds to an interest-rate shock.
- Responses are measured over several quarters.
- The result depends on model specification and shock identification.

Generate Impulse Responses

```
irf = var_result.irf(12)
irf.plot(orth=False)
plt.tight_layout()
plt.show()

This plots responses over 12 quarterly periods.
Orthogonalized responses require an identification assumption; explain it if used.
```

Read an IRF Plot

- The horizontal axis represents quarters after the shock.
- The vertical axis represents the response of a variable.
- A response may be positive, negative or close to zero.
- Confidence bands help show uncertainty around estimated responses.

Forecast Future Values

- VAR can generate forecasts using recent observations.
- Forecasts are conditional on the estimated model and available data.
- Forecasts are not guaranteed outcomes.
- Use prediction intervals and compare with a simple benchmark.

VAR Forecast: Python

```
lag_order = var_result.k_ar
forecast_input = var_stationary.values[-lag_order:]
forecast_values = var_result.forecast(
    y=forecast_input, steps=8)
forecast_index = pd.date_range(
    var_stationary.index[-1] + pd.offsets.QuarterBegin(),
    periods=8, freq='QS')
forecast = pd.DataFrame(forecast_values, index=forecast_index,
    columns=var_stationary.columns)
print(forecast)
```

Corporate Market Applications

- GDP growth can support demand and revenue planning.
- Inflation trends can inform cost and pricing discussions.
- Interest-rate movements can affect borrowing and investment costs.
- Businesses can use scenarios to support budgeting and risk planning.

Limitations of VAR Analysis

- Results depend on data quality and variable selection.
- Structural breaks may affect long historical samples.
- AIC may select a lag that is not ideal for every purpose.
- Impulse responses depend on shock-identification assumptions.
- Forecasts can become less reliable during unexpected events.

Expected Outputs and Findings

- A cleaned quarterly dataset of macroeconomic indicators.
- ADF test results for stationarity assessment.
- AIC table and selected lag length.
- VAR model summary and diagnostic results.
- Impulse-response plots and quarterly forecasts.

Conclusion

- VAR analyzes multiple economic indicators as a connected system.
- Stationarity testing supports reliable time-series modeling.
- AIC provides a systematic way to select lag length.
- Impulse responses and forecasts help explore possible market trends.
- Results should be interpreted with uncertainty and model limitations.